

Midsem

● Graded

Student

TANUSH GOEL

Total Points

129 / 150 pts

Question 1

True and False

9 / 10 pts

– 0 pts Correct

✓ – 1 pt Click here to replace this description.

– 2 pts Click here to replace this description.

– 3 pts Click here to replace this description.

– 4 pts Click here to replace this description.

Question 2

HTTP

12 / 14 pts

2.1 (no title)

3 / 3 pts

✓ - 0 pts Correct

- 3 pts Incorrect

- 1.5 pts Partially Correct

- 1 pt Mostly Correct

2.2 (no title)

3 / 3 pts

✓ - 0 pts Correct

- 3 pts Incorrect

- 1.5 pts Partially Correct

- 1 pt Mostly Correct

2.3 (no title)

2 / 3 pts

- 0 pts Correct

- 3 pts Incorrect

- 1.5 pts Partially Correct

✓ - 1 pt Mostly Correct

- 0 pts [Click here to replace this description.](#)

💬 Non-Persistent, Parallel (5 connections):
DNS lookup: 2 ms
Base object: 24 ms
First 5 objects: 24 ms (parallel)
Last 3 objects: 24 ms (parallel)
Total: $2+24+24+24 = 74$ ms.

2.4 (no title)

2 / 3 pts

- 0 pts Correct

- 3 pts Incorrect

- 1.5 pts Partially Correct

✓ - 1 pt Mostly Correct

+ 0.5 pts [Click here to replace this description.](#)

💬 Persistent, Parallel (5 connections):
DNS lookup: 2 ms
Base object: 24 ms
All 8 objects: 24 ms (1 RTT for each parallel request)
Total: $2 + 24 + 24 = 50$ ms.

2.5

(no title)

2 / 2 pts

✓ - 0 pts Correct

- 3 pts Incorrect

- 1.5 pts Partially Correct

Question 3

Socket Programming

12 / 12 pts

3.1

(no title)

3 / 3 pts

- 3 pts Description not correct

✓ - 0 pts Correct Answer

- 0.25 pts Description Incomplete.

3.2

(no title)

3 / 3 pts

✓ - 0 pts Correct

- 3 pts No Answer

- 1.5 pts Not Complete.

3.3

(no title)

3 / 3 pts

✓ - 0 pts Correct

- 3 pts Click here to replace this description.

- 2 pts Why? is wrong

3.4

(no title)

3 / 3 pts

✓ - 0 pts Correct

- 1 pt Why? is missing.

- 3 pts Click here to replace this description.

Question 4

P2P

6 / 6 pts

4.1 (no title)

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Not Attempted/Incorrect

- 1 pt Partially correct

4.2 (no title)

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Incorrect/Not attempted

4.3 (no title)

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Incorrect/Not attempted

- 1 pt Partially correct

+ 0.5 pts Click here to replace this description.

Question 5

TCP

11 / 15 pts

5.1 (no title)

5 / 5 pts

✓ - 0 pts Correct

- 5 pts Incorrect

- 5 pts Un-attempted

+ 1 pt Click here to replace this description.

5.2 (no title)

5 / 5 pts

✓ - 0 pts Correct

- 5 pts Incorrect

- 5 pts Un-attempted

5.3 (no title)

1 / 5 pts

- 0 pts Correct

✓ - 5 pts Incorrect

- 5 pts Un-attempted

+ 1 pt Click here to replace this description.

🗨 + 1 pt Point adjustment

Question 6

(no title)

24 / 24 pts

– 0 pts Correct

✓ – 8 pts Click here to replace this description.

– 14 pts Click here to replace this description.

– 12 pts Click here to replace this description.

– 10 pts Click here to replace this description.

– 4 pts Click here to replace this description.

– 22 pts Click here to replace this description.

– 24 pts Click here to replace this description.

– 6 pts Click here to replace this description.

– 18 pts Click here to replace this description.

– 20 pts Click here to replace this description.

– 2 pts Click here to replace this description.

– 16 pts Click here to replace this description.

✓ + 8 pts Click here to replace this description.

+ 6 pts Click here to replace this description.

+ 4 pts Click here to replace this description.

+ 2 pts Click here to replace this description.

– 0 pts Click here to replace this description.

Question 7
(no title)

41 / 41 pts

correct IPv4 or V6 categorization

– 3 pts Partial Correct

✓ – 0 pts All Correct

– 1 pt One mistake

– 2 pts Partially Correct

Encapsulating (Y/N)

– 3 pts Partial correct

– 6 pts Incorrect

✓ – 0 pts All Correct

– 5 pts Only one correct

– 1 pt One mistake

– 2 pts Partial correct

– 4 pts Partial correct

Source IP Address Correct Identification

– 3 pts Partial Correct (from C, c to b, b to f, a to c will have the source IP as "144.80.89.206" i.e IPV4 of C)

– 1 pt Incorrect (A to C and F to D will have the source IP - IPv6 of A)

– 5 pts Only one correct

– 2 pts Partial Correct

✓ – 0 pts All Correct

– 4 pts Partial correct

– 6 pts All Wrong

Destination IP address correct identification

– 3 pts Partial Correct (from C, c to b, b to f, a to c will have the Destination IP as "124.120.182.243" i.e IPV4 of F)

– 1 pt Partial correct (A to C, F to D will have the Destination IP as IPV6 of D)

– 5 pts Only one correct answer

– 2 pts Partial Correct

✓ – 0 pts All Correct

– 4 pts Partial Correct

– 1 pt One mistake

7.2

(no title)

7 / 7 pts

✓ - 0 pts Correct

- 0.5 pts Click here to replace this description.

- 0.5 pts Click here to replace this description.

- 2 pts Click here to replace this description.

- 4 pts Click here to replace this description.

- 7 pts Click here to replace this description.

- 1 pt Click here to replace this description.

7.3

(no title)

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Click here to replace this description.

- 1 pt Click here to replace this description.

7.4

(no title)

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Click here to replace this description.

- 1 pt Click here to replace this description.

7.5

(no title)

2 / 2 pts

✓ - 0 pts Correct

- 2 pts Click here to replace this description.

- 1 pt Click here to replace this description.

Question 8

(no title)

14 / 28 pts

8.1 (no title)

2 / 2 pts

+ 0 pts incorrect

✓ + 2 pts private

8.2 (no title)

0 / 2 pts

+ 2 pts $2^9 - 2 = 510$

✓ + 0 pts incorrect or not attempted

+ 1 pt [Click here to replace this description.](#)

8.3 (no title)

12 / 24 pts

+ 0 pts incorrect or not attempted

✓ + 2 pts A subnet 10.36.185.0/25

✓ + 2 pts B subnet 10.36.185.128/27

✓ + 2 pts C subnet 10.36.184.0/24

✓ + 2 pts A broadcast 10.36.185.127

✓ + 2 pts B broadcast 10.36.185.159

✓ + 2 pts C broadcast 10.36.184.255

+ 2 pts A start 10.36.185.1

+ 2 pts B start 10.36.185.129

+ 2 pts C start 10.36.184.1

+ 2 pts A End 10.36.185.126

+ 2 pts B end 10.36.185.158

+ 2 pts C end 10.36.184.254

+ 1 pt a end 10.36.185.127

+ 1 pt b end 10.36.185.159

+ 1 pt C end 10.36.184.255



**Indian Institute of Technology, Kanpur
Computer Science and Engineering**

**Mid Semester Exam
CS425: Computer Networks**

Instructor: Adithya Vadapalli

21/02/2025

Name: TANUSH GOEL
Roll Number: 221131

This exam contains 13 pages (including this cover page) and 8 questions. Total of marks is 150. You are allowed to bring one A4-sized cheat sheet, which is handwritten (by you!!) notes to the exam. Please write your name in ALL CAPS. You have 120 minutes to solve the exam. Good luck!

Distribution of Marks

Question	Points	Score
1	10	
2	14	
3	12	
4	6	
5	15	
6	24	
7	41	
8	28	
Total:	150	

1. This question has 10 parts. Each of them is a true or false question.

- (a) (1 point) When a new host enters a network, it uses DHCP to obtain an IP address.

True

- (b) (1 point) Since UDP does not guarantee reliable data transfer, there is no checksum in the UDP header.

False

- (c) (1 point) A student in IIT D and another student in IIT K claim that their laptop has the same IP address. It is possible that both of them are telling the truth.

True

- (d) (1 point) The primary goal of the transport layer is to transport packets from one host to another. process
↓

False

- (e) (1 point) Recursive DNS is the most preferred type of DNS resolution.

False

- (f) (1 point) While doing socket programming in UDP, it is not required to specify the send function's IP address.

~~True~~ False

- (g) (1 point) The only purpose of cookies is to track user behaviour.

False

- (h) (1 point) Since NAT uses port numbers, it resides in the transport layer.

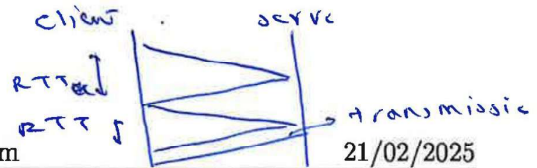
False

- (i) (1 point) Network Layer ensures that packets are transported reliably over an unreliable channel.

True

- (j) (1 point) While doing socket programming in TCP, specifying the IP address in the send function is not required.

~~False~~ True



2. Suppose you click on a link and obtain a web page within a web browser. Also, suppose that the IP address for the associated URL is not yet cached in your localhost. Therefore, a DNS lookup becomes necessary to obtain the IP address. Suppose that only one DNS server, the local DNS cache, is visited with an an RTT delay of $RTT_0 = 2$ ms. Initially, let's suppose that the Web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Suppose the RTT between the local host and the Web server containing the object is $RTT_{HTTP} = 12$ ms.

- (a) (3 points) Assuming zero transmission time for the HTML object, how much time (in msec) elapses from when the client clicks on the link until the client receives the object?

~~Given~~ The first step is to obtain the IP address using the local DNS server. The RTT for such a lookup is given as 2ms. (RTT_0)

Now we have the IP address. The localhost queries the web server for the object which has a total delay of $2RTT_{HTTP}$ + transmission time (given as 0)

Hence ans : $(RTT_0 + 2RTT_{HTTP}) = \boxed{26\text{ms}}$

- (b) (3 points) Now suppose the HTML object references 8 very small objects on the same server. Neglecting transmission times, how much time (in msec) elapses from when the client clicks on the link until the base object and all 8 additional objects are received from web server at the client, assuming non-persistent HTTP and no parallel TCP connections?

Given non-persistent non parallel TCP connection.
The 1st time is for the lookup from local : RTT_0
DNS

Then for each object request the time required would be ~~2RTT~~ $2RTT_{HTTP}$ (transmission time is 0)

Total $\Rightarrow$ 9 objects [1 main and 8 reference]

Total time : $RTT_0 + 9(2RTT_{HTTP})$
 $\Rightarrow 2\text{ms} + (18)(12)\text{ms} = \boxed{218\text{ms}}$

- (c) (3 points) Suppose the HTML object references 8 very small objects on the same server, but assume that the client is configured to support a maximum of 5 parallel TCP connections, with non-persistent HTTP.

~~RTT₀ is common. A total of 9 objects can be brought in 2 packets~~

There are again a total of 9 objects. The first 5 can be brought together using the 5 TCP (parallel) connections and the remaining 4 can be brought in the next TCP connection.

$$\begin{aligned}
 \text{Hence the total time} &= RTT_0 + 2 \underbrace{(RTT_{\text{HTTP}})}_{\text{for the first 5}} + \underbrace{2(RTT_{\text{HTTP}})}_{\text{for the new 4}} \\
 &= 2\text{ms} + (12\text{ms})4 \\
 &= \boxed{50\text{ms}}
 \end{aligned}$$

- (d) (3 points) Suppose the HTML object references 8 very small objects on the same server, but assume that the client is configured to support a maximum of 5 parallel TCP connections, with persistent HTTP.

With persistent HTTP, the greeting has to be made only 1 time. RTT_0 is same.

For the first 5 objects time = $2RTT_{\text{HTTP}}$

and for the next 4 time = RTT_{HTTP} (over the same TCP link)

$$\begin{aligned}
 \text{Hence total time} &= RTT_0 + 3RTT_{\text{HTTP}} \\
 &= 2 + 3(12) = \boxed{38\text{ms}}
 \end{aligned}$$

- (e) (2 points) What is the fastest method we have explored: Nonpersistent-serial, Nonpersistent-parallel, or Persistent-parallel?

It's definitely persistent parallel ~~HTTP~~ ^{TCP}.

Persistent prevents repeated welcoming of the client and establishes a one-time access path.

Parallel allows for multiple objects to be retrieved parallelly.

Hence outperforming the others.

⊛ does not take in IP as a parameter.

CS425

Mid Semester Exam

21/02/2025

3. In this question, we will explore the differences between UDP socket programming and TCP socket programming.

(a) (3 points) Recall that in TCP server there is a function call to `listen()`, while in UDP there is no such call. Why?

The TCP server listens for the incoming clients and welcomes using a common socket. The new TCP socket for the server is different from the welcoming socket. This is not the case in UDP where direct connect() and recv() establishes a client-server

(b) (3 points) While sending messages in the UDP, IP address is a parameter. However, that is not the case in TCP. Why?

IA UDP → destination. The welcome port or the listen port establishes a new ~~client socket~~ when the client tries to connect, and ~~uses this socket~~ for communication. This ~~client socket~~ carries the client's IP and the server's IP and hence ~~can just be specific~~ using the port notes.

(c) (3 points) Can a single socket be used to receive messages from multiple IP addresses in TCP? Why?

⊛ No: a socket is governed by the 4 tuple (sender IP, sender port, receiver IP, receiver port). Hence for different sender address IP there would be different sockets.

(d) (3 points) Can a single socket be used to receive messages from multiple IP addresses in UDP? Why?

In UDP the socket is defined by the tuple (destination IP, destination port)

Hence even if the sender IPs are different the socket might be the same.

Ans: Yes

4. In this problem, you will compare the time needed to distribute a file that is initially located at a server to clients via either client-server download or peer-to-peer download. The problem is to distribute a file of size $F = 9$ Gbits to each of these 6 peers. Suppose the server has an upload rate of $u = 79$ Mbps. The 6 peers have upload and download rates of:

1. $u_1 = 18$ Mbps, $d_1 = 29$ Mbps (Client 1)
2. $u_2 = 11$ Mbps, $d_2 = 27$ Mbps (Client 2)
3. $u_3 = 14$ Mbps, $d_3 = 16$ Mbps (Client 3)
4. $u_4 = 19$ Mbps, $d_4 = 39$ Mbps (Client 4)
5. $u_5 = 18$ Mbps, $d_5 = 18$ Mbps (Client 5)
6. $u_6 = 12$ Mbps, $d_6 = 23$ Mbps (Client 6)

(a) (2 points) What is the minimum time needed to distribute this file from the central server to the 6 peers using the client-server model?

Ans: 683.5 sec

client uploading time = $\left(\frac{NF}{U}\right) = \frac{6 \times 9 \text{ Gb}}{79 \text{ Mb}} = \frac{54000}{79} \text{ s}$

minimum downloading amongst clients = $d_3 = 16 \text{ Mbps} = \frac{54000}{16} \text{ s}$

time taken to download using $d_3 = F/d_3 = \frac{9 \text{ Gb}}{16 \text{ Mbps}}$

Ans: $\max(NF/U, F/d_3) = \frac{54000}{16} \text{ s}$

(b) (2 points) For the previous question, which of the following is the root cause of this specific minimum time? Server, Client 1, Client 2, Client 3, Client 4, Client 5, or Client 6?

Server

(c) (2 points) What is the minimum time needed to distribute this file using peer-to-peer download?

minimum time needed = $\max\left(\frac{F}{U}, \frac{F}{d_3}, \frac{NF}{U + \sum u_i}\right)$ where $d_3 = \text{minimum download rate}$

$F/U = (9 \text{ Gb}) / 79 \text{ Mb}$

$F/d_3 = 9 \text{ Gb} / 16 \text{ Mb}$

$\frac{U + \sum u_i}{N} = \frac{79 + 18 + 11 + 14 + 19 + 18 + 12}{6} = \frac{171}{6} = 28.5 \text{ Mbps}$

Hence the maximum would be = $\frac{F}{d_3} = \frac{9 + 10^3}{16} \text{ s}$

$= \frac{(9 + 125)}{2} \text{ s}$

$= 562.5 \text{ s}$

5. Consider the figure (in the next page) in which a TCP sender and receiver communicate over a connection in which the segments can be lost. The TCP sender wants to send a total of 10 segments to the receiver and sends an initial window of 5 segments at $t = 1, 2, 3, 4$, and 5 , respectively. Suppose the initial value of the sequence number is 180 and every segment sent to the receiver each contains 408 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at $t = 8$, and an ACK for this segment arrives at $t = 15$. As shown in the figure, 1 of the 5 segments is lost between the sender and the receiver, but none of the ACKs are lost. Assume there are no timeouts and any out-of-order segments received are thrown out.

- (a) (5 points) What is the sequence number of the segment sent at T1, T2, T3, T4, and T5?

~~sequence no.~~

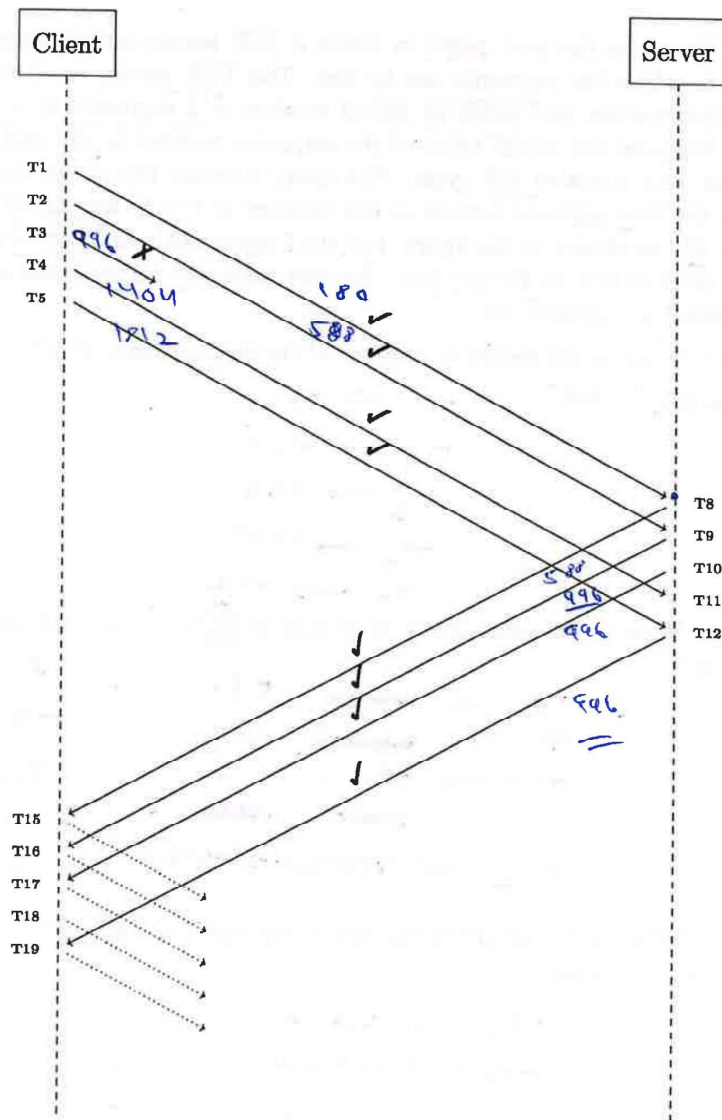
$T_1 \rightarrow 180$
 $T_2 \rightarrow 588$
 $T_3 \rightarrow 996$
 $T_4 \rightarrow 1404$
 $T_5 \rightarrow 1812$

- (b) (5 points) What is the value of the ACK sent at T8, T9, T10, T11, and T12? (If segment lost, write 'x')

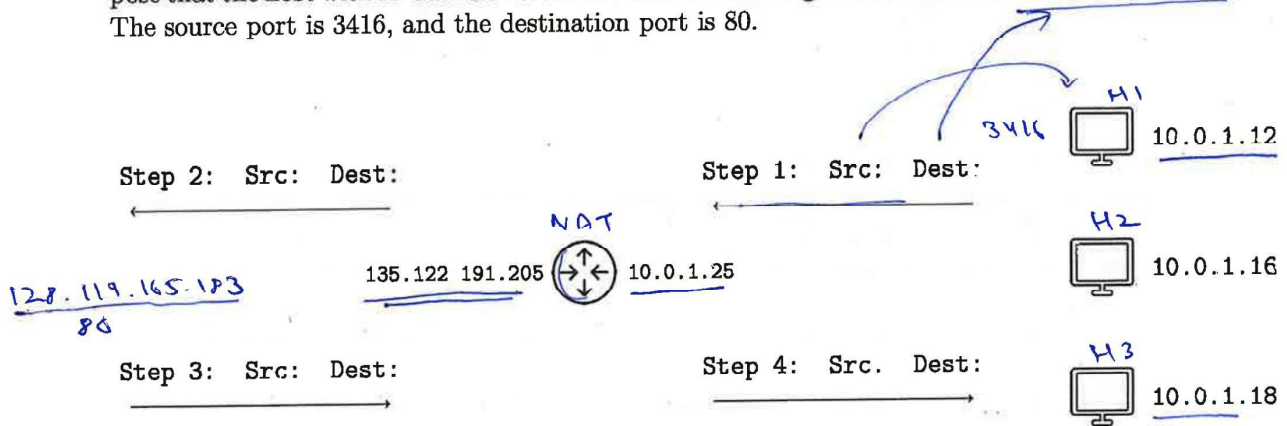
$T_8 \rightarrow 588$
 $T_9 \rightarrow 996$
 $T_{10} \rightarrow X$
 $T_{11} \rightarrow 996$
 $T_{12} \rightarrow 996$

- (c) (5 points) What is the sequence number of the segment sent at T15, T16, T17, T18, and T19? (If segment lost, write 'x')

$T_{15} \rightarrow 588$
 $T_{16} \rightarrow 996$
 $T_{17} \rightarrow X$
 $T_{18} \rightarrow 996$
 $T_{19} \rightarrow 996$

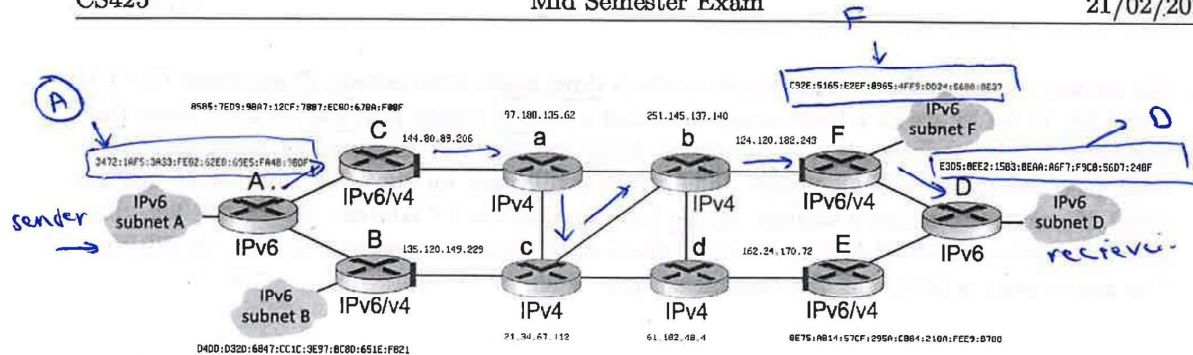


6. (24 points) Consider the scenario below in which three hosts, with private IP addresses 10.0.1.12, 10.0.1.16, 10.0.1.18 are in a local network behind a NAT'd router that sits between these three hosts and the larger Internet. IP datagrams being sent from, or destined to, these three hosts must pass through this NAT router. The router's interface on the LAN side has IP address 10.0.1.25, while the router's address on the Internet side has IP address 135.122.191.205. Suppose that the host with IP address 10.0.1.12 sends an IP datagram destined to host 128.119.165.183. The source port is 3416, and the destination port is 80.



Complete the following table:

Step #	Source IP Address	Destination IP Address	Src Port Change? (Yes / No)
Step 1	10.0.1.12	128.119.165.183	No
Step 2	135.122.191.205	128.119.165.183	Yes
Step 3	128.119.165.183	135.122.191.205	No
Step 4	128.119.165.183	10.0.1.12	Yes No



7. Consider the network shown below which contains four IPv6 subnets, connected by a mix of IPv6-only routers, IPv4-only routers and dual-capable IPv6/IPv4 routers. Suppose that a host of subnet A wants to send an IPv6 datagram to a host on subnet D. Assume that the forwarding between these two hosts goes along the path: $A \rightarrow C \rightarrow a \rightarrow c \rightarrow b \rightarrow F \rightarrow D$.

(a) (28 points) Fill in the following table: *F, A and D have been circled in the diagram*

Datagram	IPv4 or v6	Encapsulating (Y/N)	Source IP Address	Destination IP Address
A to C	v6	N	A	D
from C to a	v4	Y	144.80.89.206	124.120.182.243
c to b	v4	Y	144.80.89.206	124.120.182.243
b to F	v4	Y	144.80.89.206	124.120.182.243
F to D	v6	N	A	D
a to c	v4	Y	144.80.89.206	124.120.182.243
F to D	v6	N	A	D

- (b) (7 points) In the above question, for all the encapsulated packets is encapsulated, what are the Source and Destination Addresses (of the encapsulated packets)? If there is no encapsulation in any of the below datagrams in the table, just write "NA".

Datagram	Source IP Address	Destination IP Address
A to C	NA	NA
from C	A	D
c to b	A	D
b to F	A	D
F to D	NA	NA
a to c	A	D
F to D	NA	NA

A = 3472 : 1A F5 :
 3A 33 : FE 22 :
 12 EB : 69 E5 :
 FA 48 : 9B DF
 D = E 305 : BEE2 :
 15 B3 : BE AA :
 A 147 : F9 6
 56 07 : 24 8F

- (c) (2 points) What router is the tunnel entrance? Give the router's letter

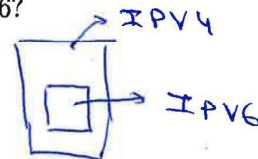
Router C

- (d) (2 points) What router is the tunnel exit? Give the router's letter

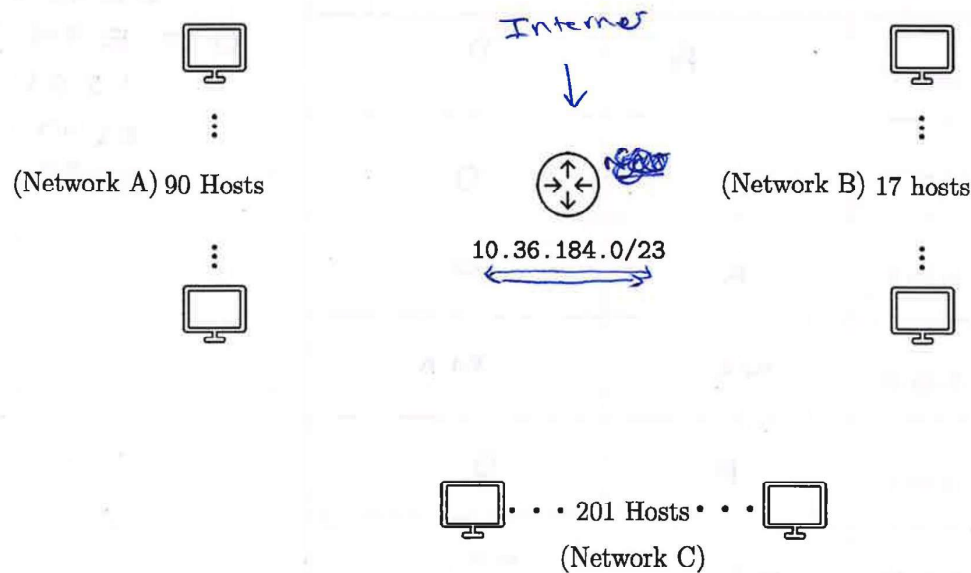
Router F

- (e) (2 points) Which protocol encapsulates the other, IPv4 or IPv6?

IPv4 encapsulates IPv6



8. Consider the router and the three attached subnets below (A, B, and C). The number of hosts is also shown below. The subnets share the 23 high-order bits of the address space: 10.36.184.0/23. Assign subnet addresses to each of the subnets (A, B, and C) so that the amount of address space assigned is minimal, and at the same time leaving the largest possible contiguous address space available for assignment if a new subnet were to be added. Then answer the questions below. (*Hint: The last address in the subnet is the broadcast address and the first address is the subnet ID!*)



- (a) (2 points) Is the address space public or private?

private

- (b) (2 points) How many hosts can there be in this address space?

There are 9 bits which can ~~be~~ point to a host ($32 - 23 = 9$) that can be accommodated.
Hence the total hosts are = 512 hosts

(c) (24 points) Fill in the following table:

Network	Subnet Addr (CIDR)	Broadcast Address	Starting Address	Ending Address
A	10.36.185.0/25	10.36.185.127	10.36.185.0	10.36.185.255
B	10.36.185.128/27	10.36.185.159	10.36.185.128	10.36.185.191
C	10.36.184.0/24	10.36.184.255 10.36.184.0 10.36.184.255	<u>10.36.184.0</u>	<u>10.36.184.255</u>

C :

(201) : 8 bit

A = (90) : 7 bits
are enough

B : (17) : 5 bits

C : 201 hosts $\Rightarrow$ 8 bit : 10.36.184.0/24
~~10.36.184.0~~A : 90 hosts $\Rightarrow$ 7 bit : 10.36.185.0/25B : 17 hosts $\Rightarrow$ 5 bit :

